

1 The structure of an argument

Booth, Colomb, Williams, Bizup, and Fitzgerald treat a research argument as a conversation rather than a proof. Part III of *The Craft of Research* (4th edition, University of Chicago Press, 2016) sets out five elements and the reader questions each one answers. The frame is useful twice over. It is a checklist for a draft, and it is a diagnostic for a review, because a reviewer's objection is always an objection to one of the five.

1.1 The five elements

Chapter 7 opens with the summary sentence: "In a research argument, you make a claim, back it with reasons supported by evidence, acknowledge and respond to other views, and sometimes explain your principles of reasoning" (7.1, p. 110). The authors then derive the five elements from the questions a skeptical listener asks (7.1, p. 111):

1. **Claim.** What do you want me to believe? What's your point?
2. **Reasons.** Why do you say that? Why should I agree?
3. **Evidence.** How do you know? Can you back it up?
4. **Acknowledgment and response.** But what about ...?
5. **Warrant.** How does that follow? What's your logic? Can you explain your reasoning?

The definitions are given in a boxed note on p. 112. A claim is "an assertion (which can be a single sentence or more) that demands support." A reason is "an assertion that supports a claim." Evidence is "data deployed to support a reason," and, unlike the other two, evidence is not always framed as an assertion. The core of the argument is then written as a single line: claim because of reason based on evidence (7.2.2, p. 114).

Two properties of a claim are assessed separately from its support. The claim must be specific, and it must be significant. Significance is defined operationally: it is "a quality they measure by how much it asks them to change what they think," and the estimate is the number of other beliefs a reader would have to revise on accepting the claim (8.2.2, p. 127).

1.2 Reasons are not evidence

The distinction is the one writers most often collapse, and the book spends a full section on it because collapsing it produces arguments that look supported and are not.

The authors concede that the two words are interchangeable in some contexts. Their test is a pair of sentences (7.2.2, p. 113). "What evidence do you base your reason on?" is well formed. "What reason do you base your evidence on?" is not. Two further differences follow. "We use our minds to think up reasons." Evidence has to be searched for "out there" in the world and then made available for everyone to see. Hence the diagnostic: "It makes no sense to ask, Where do I go to see your reasons? It does make sense to ask, Where do I go to see your evidence?"

The reason writers conflate them is that a chain of reasons feels like support. A reason is also a subclaim, and a subclaim can itself take reasons, so the chain can run for some distance without ever touching data. The authors state the stopping condition directly: these chains do not go on forever, and eventually you have to show some data.

Where the chain stops is not the writer's decision. Chapter 9 makes the boundary a property of the audience: "To count as evidence, a statement must report something that readers agree not to question, at least for the purposes of the argument" (9.2, p. 133). If readers do question it, what the writer took for hard evidence is only a soft reason, and the argument has not reached bottom. A sidebar on the same page puts it as a rule: evidence counts as evidence only if readers accept it without question, at least for the moment.

Chapter 9 adds a complication that matters for empirical work. A paper almost never contains the evidence. It contains a report of the evidence: a table, a figure, a summary statistic. Data taken from a source have already been shaped by that source, and reporting them again shapes them once more, if only by placing them in a new context. The recommendation is to state how the data were collected and by whom, and to cite a source as close to the evidence as possible.

The criteria for evidence are a list of five (9.4, p. 138). Readers want evidence to be accurate, precise, sufficient, representative, and authoritative. Relevance is treated separately, because relevance is a question about the warrant.

1.3 Warrants

A warrant is "a principle that connects a reason to a claim" (chapter 11, p. 155). It exists because a reader can grant that a reason is true and still deny that it is relevant. The authors' running example is a reader who accepts that the line of hard freezes is moving north, accepts that health care costs are rising, and asks what the first has to do with the second (7.4, p. 115).

A warrant has two parts, a general circumstance and a general consequence, and its logic is that if a generalization holds, so must specific instances of it. For the inference to go through, a reader has to accept four things: that the warrant is reasonable, that the reason is true, that the specific circumstance in the reason is a plausible instance of the general circumstance, and that the specific consequence in the claim is a plausible instance of the general consequence (7.4, p. 116). Most academic warrants are stated in compressed form, with circumstance and consequence run together, but both parts can always be recovered.

When to state one. The governing rule appears in the chapter abstract and again in its first page: "state your warrants only if your readers will not be able to understand your reasoning without them, or if you anticipate that your reasoning will be challenged" (chapter 11, p. 155). Section 11.5 (p. 164) names three occasions. The readers are outside the field. The principle of reasoning is new or controversial within the field. The claim is one readers will resist because they do not want it to be true, in which case stating a warrant they already accept, before the reason and claim, is the recommended order.

The converse is equally explicit. Experienced researchers writing for specialists leave most warrants unstated, because stating the obvious reads as condescension rather than help (11.2, p. 158). This is why arguments written for specialists are hard for newcomers to follow: the connecting principle is present and invisible.

Five tests. Before a reader challenges a warrant, the writer should ask whether it is reasonable,

whether it is sufficiently limited, whether it is superior to competing warrants, whether it is appropriate to this field, and whether it is able to cover the reason and claim (11.4, p. 161). The last test is the one that catches bad arguments that look valid. The authors work an example on violent video games in which the warrant is acceptable, the reason is true, and the argument still fails, because the stated reason is not an instance of the warrant's general circumstance (11.6, p. 166).

Prefer evidence. The closing Quick Tip warns against building on warrants when data are available. A claim of fact cannot be supported by a warrant and a reason alone. The instruction is to rely “not on elaborate lines of reasoning based on warrants but on hard evidence” (Quick Tip, p. 172).

1.4 Acknowledgment and response

This is the element that separates an argument from an assertion. The chapter 10 abstract states it as a condition: “An argument is not complete if it fails to recognize other points of view” (p. 141). Chapter 7 says the same thing about the diagram: no research argument is complete without acknowledgments and responses (7.3, p. 115). A text that offers only claims, reasons, and evidence may still read as thin, or as ignorant of or dismissive of the views it did not name.

The authors sort anticipated objections into two kinds (p. 142). Objections to **intrinsic** soundness challenge the clarity of the claim, the relevance of the reasons, or the quality of the evidence. Objections to **extrinsic** soundness ask the writer to consider alternatives: a different framing of the problem, evidence that was overlooked, or what others have written. The second kind is usually found by reading sources for disagreement rather than for data, including disagreements among the sources themselves (10.2, p. 145).

Section 10.1 (pp. 143 to 144) lists the questions a hostile reader asks of the support. One asks for a different kind of evidence. The rest attack quality: it is not accurate, not precise enough, not current, not representative, not authoritative. The authors identify the seventh as the hardest to answer, and it is the one that asks for more of it.

A response is not a competing claim. Even a minimal response supplies a reason for limiting or rejecting what was acknowledged, and a substantial one is a subordinate argument with its own reasons, evidence, and possibly its own warrant (10.4, p. 148).

Two further points are worth carrying into a rebuttal. First, acknowledging a flaw that cannot be fixed is recommended over concealing it, with three usable forms of response: that the rest of the argument outweighs the flaw, that the flaw is serious but further research will find a way around it, or that the flaw prevents full acceptance of the claim while the argument still shows what a better answer would need (10.3.2, p. 147). Second, on how to phrase disagreement: “Save criticism for the response, and direct it at the work rather than the person” (10.5.1, p. 150).

Selection is a real constraint. Section 10.3 (p. 145) calls it a Goldilocks problem. Too many acknowledgments distract from the core; too few make the writer seem indifferent to the reader's views. The priority list is plausible charges the writer can rebut, alternative lines of argument important in the field, alternative conclusions readers want to be true, alternative evidence readers already know, and counterexamples that have to be addressed.

1.5 Testing an argument by asking a reader's questions

The book's method for testing a draft is to run the five questions against it before a reader does. Section 10.1 gives the procedure. Write the core first, then question it as someone with a stake in a different outcome would. The suggested posture is not warfare but a warm debate, adopted on behalf of a reader who wants the argument to be wrong.

The questions come in two batches. The first is about the problem and the solution: whether there is a problem at all, whether it has been properly defined, whether the solution is practical or conceptual and whether it matches the kind of problem, whether the claim is stated too strongly, and why this solution is better than the alternatives. The second is about the support, and is the list of evidence objections above. The instruction attached to the first batch is worth quoting for its cost logic: the time to fix a problem with an argument is when you find it.

The Quick Tip after chapter 10 (pp. 153 to 154) adds three disagreements that at least some readers always produce. There are causes in addition to the one claimed. There are counterexamples. And, third, "I don't define X as you do." On the last, the authors note that definitions function as warrants, so a dispute over a definition is a dispute over the connection between a reason and a claim, not a matter of vocabulary.

1.6 Reading a reviewer comment as an attack on one element

A review is the reader's half of the conversation, delivered after the fact. Every substantive objection lands on one of five places. Identifying which one is faster than parsing the comment sentence by sentence, and it determines what a response can consist of.

Target	Form of the objection	What the response must supply
Claim	Overstated, vague, or not significant	A narrower or more precise claim, or an argument that acceptance would change what readers believe
Evidence	Not accurate, precise, sufficient, representative, or authoritative	Data, or an account of collection, or a stated limit
Warrant	The reason is granted but read as irrelevant	The connecting principle, stated, and a check that reason and claim are instances of it
Definition	A term is used in a sense the reader does not hold	The definition, with a subordinate argument for it
Significance	So what	The consequence of the result for what readers currently think or do

Three consequences follow.

An objection to the warrant cannot be answered with more evidence. The reviewer has already granted the reason. Adding a table addresses a question that was not asked. The response has to state the principle that makes the reason relevant, then show that the reason and the claim

are good instances of it. If neither can be shown, the argument needs revision rather than defense (11.6, p. 168).

An objection to the evidence cannot be answered with more reasoning. This is the same rule read in the other direction. A claim of fact does not follow from a warrant and a reason. If a reviewer says the evaluation is thin, the answer is a run, a number, or a stated limit.

An objection to a definition is an objection to a warrant. Reviewers frequently write that a word in the paper does not mean what the paper takes it to mean. Treating that as a terminology complaint understates it. The definition licenses the inference, so the response owes a definition and a short argument for choosing it, including an acknowledgment of the common or competing sense and a reason for not using it.

One more use of the frame. When the same objection is raised by two reviewers in different words, sorting both by target shows whether it is one objection or two. Two comments that both target the evidence can be answered once. A comment on the evidence and a comment on the warrant cannot, however similar their wording.

1.7 Sources

- Wayne C. Booth, Gregory G. Colomb, Joseph M. Williams, Joseph Bizup, and William T. FitzGerald, *The Craft of Research*, 4th edition, University of Chicago Press, 2016. ISBN 978-0-226-23973-6. Every quotation and page reference above was read from the copy at [_side/_tips/the-craft-of-research-4th.pdf](#). Page numbers are the printed page numbers in the running heads, not the page numbers of the file.
- Sections used: chapter 7, *Making Good Arguments: An Overview*, pp. 110 to 121; chapter 8, *Making Claims*, pp. 122 to 131; chapter 9, *Assembling Reasons and Evidence*, pp. 132 to 140; chapter 10, *Acknowledgments and Responses*, pp. 141 to 152, with the Quick Tip *Three Predictable Disagreements*, pp. 153 to 154; chapter 11, *Warrants*, pp. 155 to 170, with the Quick Tip *Reasons, Evidence, and Warrants*, pp. 171 to 172.

The table and the three consequences in the previous subsection are not in the book. They apply the book's five elements to reviewer comments, which is an extension the authors do not make.